Amrita Mazumdar

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Present Employment

2026-Now Senior Research Scientist, NVIDIA Research
Researcher with interests in compressed representations of 2D, 3D, and 4D scenes. Manager: Shalini De Mello

2021-2025 Research Scientist, NVIDIA Research

Education

University of Washington

June 2020 *PhD in Computer Science & Engineering*

March 2017 *Masters of Computer Science & Engineering*

Advisors: Luis Ceze & Mark Oskin

Columbia University

May 2014 *Bachelor of Science in Computer Engineering*

Publications

2026 *Grounded-Exo2Ego: Structured Semantic Grounding for Robust Exocentric-to-Egocentric Video Generation*
S. Wang, M. Stengel, T. Li, S. Park, A. Mazumdar, K. Nagano, A. Trevithick, S. De Mello
NeurIPS 2026.

World from Motion: Generative Dynamic Gaussian Reconstruction from Monocular Video
L. Zhu, S. Huang, A. Mazumdar, T. Li, Z. Gojcic, G. Wetzstein, I. Armeni, S. De Mello, A. Trevithick
NeurIPS 2026.

DyaPlex: Full-Duplex Speech-Motion Model for Dyadic Interaction
K. Nagano, H. Liu, S. Park, T. Li, A. Mazumdar, C. Jacobsen, S. Wang, M. Stengel, R. Roy, K. C. Cheung, S. See, S. De Mello
NeurIPS 2026.

NemotronLabs VoiceChat: An Open Full-duplex Speech-to-Speech Model with Tool Calling Capabilities
J. Balam, T. M. Bartley, E. Casanova, S. S. Chauhan, C. Chen, Z. Chen, Z. Chen, S. Deng, S. De Mello, M. T. Desta, H. Dubey, S. Essid, N. Ferchichi, B. Ginsburg, M. Graterol Fuenmayor, K. Hu, M. Kim, V. Klimkov, S. Lakshmi Narasimhan, Y.-H. Lee, J. Li, E. M. Peters Long, A. S. Mahabaleshwarkar, A. Malte, A. Mazumdar, V. Mendelev, K. Nagano, O. Olabiyi, S. Park, A. Pasad, Y. Peng, E. Rastorgueva, R. Roy, N. Srihari, Y. Su, Y. Suhara, V. A. Trinh, J. Wang, P. Zelasko
Workshop on Real-Time Multimodal Conversational AI, NeurIPS, Paris, France, December 9–13, 2026.

VideoFDB: Evaluating Full-Duplex Vision-Speech Capabilities in Conversational Agents
A. Mazumdar, S. Park, R. Roy, N. Srihari, S. Wang, Y. Zhou, J. Wang, K. Nagano, S. De Mello
arXiv preprint, 2026.

TSMC: Time-Varying Scene Mesh Compression.
G. Chen, L. Váša, A. Mazumdar, M. Dasari.
SIGGRAPH 2026.

Real-Time Neural Filtering for Perceptual Enhancement of Digital Experiences
A. Patney, S. Kumar, D. Brittain, A. Russell, E. Xu, A. Mazumdar, A. Bernal
SID Display Week 2026, Invited paper.

2025 *Play4D: Accelerated and Interactive Free-viewpoint Video Streaming for Virtual Reality and Light Field Displays*
J. Kim, M. Stengel, A. Mazumdar, T. Li, C. Sun, D. Luebke, S. De Mello
SIGGRAPH Asia Emerging Technologies.

Democratizing Immersive Experiences with NVIDIA AI
A. Mazumdar*, T. Li*, M. Stengel, J. Kim, S. De Mello, W. Braithwaite, C. Sun, S. Park
NVIDIA GTC 2025.

2024 *QUEEN: QUantized Efficient ENcoding for Streaming Free-viewpoint Videos*
S. Girish, T. Li, A. Mazumdar, A. Shrivastava, D. Luebke, S. De Mello
NeurIPS 2024.

GRACE: Loss-Resilient Real-Time Video through Neural Codecs
Y. Cheng, Z. Zhang, H. Li, A. Arapin, Y. Zhang, Q. Zhang, Y. Liu, X. Zhang, F. Y. Yan, A. Mazumdar, N. Feamster, and J. Jiang
NSDI 2024.

2023 *Optimizing Real-Time Video Experience with Data Scalable Codec*
H. Li, Y. Cheng, Z. Zhang, Q. Zhang, A. Arapin, N. Feamster, A. Mazumdar
SIGCOMM EMS 2023.

Online Overexposed Pixels Hallucination in Videos with Adaptive Reference Frame Selection
Y. Xing, A. Mazumdar, A. Patney, C. Liu, H. Yin, Q. Chen, J. Kautz, I. Frosio
https://arxiv.org/abs/2308.15462.

AI-Mediated 3D Video Conferencing
M. Stengel, K. Nagano, C. Liu, M. Chan, A. Trevithick, S. De Mello, J. Kim, D. Luebke
Additional Contributors: A. Mazumdar, S. Wang, M. Jaiswal
SIGGRAPH Emerging Technologies.

2021 *Vss: A storage system for video analytics*
B. Haynes, M. Daum, D. He, A. Mazumdar, M. Balazinska, A. Cheung, L. Ceze
SIGMOD 2021.

TASM: A Tile-Based Storage Manager for Video Analytics
M. Daum, B. Haynes, D. He, A. Mazumdar, M. Balazinska

ICDE 2021.

2020 *VisualWorldDB: A DBMS for the Visual World*
B. Haynes, M. Daum, A. Mazumdar, M. Balazinska, L. Ceze, A. Cheung, M. Oskin
CIDR, 2020.

2019 *Vignette: Perceptual Compression for Video Storage and Processing Systems*
A. Mazumdar, B. Haynes, M. Balazinska, L. Ceze, A. Cheung, M. Oskin
ACM Symposium on Cloud Computing (SoCC), 2019.
Best Poster Award Winner.

Visual Road: A Video Data Management Benchmark
B. Haynes, A. Mazumdar, M. Balazinska, L. Ceze, A. Cheung
SIGMOD 2019.

2018 *LightDB: A DBMS for Virtual Reality Video*
B. Haynes, A. Mazumdar, A. Alaghi, M. Balazinska, L. Ceze, A. Cheung
Proceedings of the VLDB Endowment (PVLDB) 11(10), 2018.

Application Codesign of Near-Data Processing for Similarity Search
V. T. Lee, A. Mazumdar, C. del Mundo, A. Alaghi, L. Ceze, M. Oskin
IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2018.

2017 *Exploring In-Camera Computation-Communication Tradeoffs*
A. Mazumdar, T. Moreau, S. Kim, A. Alaghi, L. Ceze, M. Oskin, and V. Sathe
IEEE International Symposium on Workload Characterization (IISWC), 2017.

A Hardware-Friendly Bilateral Solver for Real-Time Virtual Reality Video
A. Mazumdar, A. Alaghi, J. T. Barron, D. Gallup, L. Ceze, M. Oskin, and S. M. Seitz
High-Performance Graphics (HPG), 2017.

2010 *Principles and Techniques of Schlieren Imaging Systems*
A. Mazumdar
Technical Report CUCS-016-13, 2010.

Patents

2026 US Patent 12,711,703: *Techniques for selection and inclusion of virtual assets in a virtual scene.* A. Mazumdar, J. A. D. Cameron, K. I. Shkurko, R. D. Lyle, P. Mahapatra.
US Patent App. 19/181,188: *Generalizable learned triplane compression.* A. Mazumdar, S. De Mello.

2025 US Patent 12,452,542: *Hallucinating details for over-exposed pixels in videos using learned reference frame selection.* I. Frosio, Y. Xing, C. Liu, A. Patney, H. Yin, A. Mazumdar, J. Kautz.
US Patent App. 19/194,581: *Frameworks for implementing streamable and hardware accelerated neural 3d volumes.* M. Stengel, A. Mazumdar, K. Nagano, M. Chan, S. De Mello, A. Evans, D. Luebke.
US Patent App. 19/194,579: *Frameworks for implementing streamable and hardware accelerated neural 3d volumes.* M. Stengel, A. Mazumdar, K. Nagano, M. Chan, S. De Mello, A. Evans, D. Luebke.

US Patent App. 19/194,585: *Frameworks for implementing streamable and hardware accelerated neural 3d volumes.* M. Stengel, A. Mazumdar, K. Nagano, M. Chan, S. De Mello, A. Evans, D. Luebke.
US Patent App. 19/180,743: *Generalizable learned triplane compression.* A. Mazumdar, S. De Mello.
US Patent App. 19/016,225: *Quantized efficient encoding for streaming free-viewpoint videos.* A. Mazumdar, T. Li, S. De Mello, S. Girish.
US Patent App. 18/800,563: *Model-based processing to reduce reaction times for content streaming systems and applications.* A. Patney, A. Mazumdar, A. I. Russell, K. Gnanasekaran, S. Schneider, R. Brown, R. Tarrazo, S. Goyal, A. Banerjee, J. Mawdsley.
US Patent App. 18/598,777: *Saliency-guided packet loss mitigation for content streaming systems and applications.* Y. Su, A. Mazumdar, A. Bosnjakovic, J. Zimmermann.
US Patent App. 18/532,890: *Techniques for selection and inclusion of virtual assets in a virtual scene.* A. Mazumdar, J. A. D. Cameron, K. I. Shkurko, R. D. Lyle, P. Mahapatra.
2023 US Patent 11,729,407: *Saliency-based video compression systems and methods.* A. Mazumdar, L. Ceze, M. Oskin.

Honors & Awards

2025 CVPR Outstanding Reviewer (Top 5%)
2020 UW CoMotion Commercialization Fellowship
UW CoMotion STEP Award Winner
2019 UW CoMotion Innovation Gap Fund Winner
SoCC Best Poster Award, for the poster accompanying the SoCC 2019 paper
2018 NCWIT Collegiate Award Finalist
2017 UW Allen School Madrona Prize Runner-Up
2014 Google Anita Borg Memorial Fellowship
2013 CRA-W Distributed Research Experience for Undergraduates (DREU) Fellowship

Software & Product Impact

2026 *VideoFDB dataset* — released on HuggingFace (2.5k downloads).
2024 *RTX Digital Vibrance* — released in the NVIDIA App.
QUEEN — NVIDIA open source release (187 GitHub stars). Demos with NVIDIA XR at GTC 2025 in San Jose and Paris and SIGGRAPH 2025.
2018 *LightDB* — database management system for virtual reality video (34 GitHub stars).
2017 *HFBS* — software and hardware implementations of a bilateral solver (27 GitHub stars).

Service

Reviewer: *CVPR, ECCV, WACV, NeurIPS, AAAI, SIGGRAPH, SIGGRAPH Asia, Transactions on Graphics, SIGCOMM, ISCA, MICRO, IEEE Micro, PACT, IISWC.*
Program Chair/Co-Chair: *CVPR IPA* (2026), *SIGCOMM EMS* (2025).
Artifact Evaluation Committee: *OSDI* (2021), *ASPLOS* (2021).

Institutional Service

2025-Now

Member, *NVIDIA Academic Grant Review Committee*
2022-Now Member, *NVIDIA Graduate Fellowship Review Committee*
2017-20 Member, *University of Washington CSE Graduate Admissions Committee*
2015-20 Member, *University of Washington CSE Prospective Student Committee*
2016 Co-Chair, *University of Washington CSE Prospective Student Committee*
Co-Chair, *University of Washington CSE TGIF Committee*
2015 Mentor, *University of Washington CSE Graduate Mentoring*
Member, *University of Washington CSE Prospective Student Committee*

Tutorials and Workshops Organized

2026 Workshop on Interactive Physical AI (IPA), *with CVPR 2026.*
Workshop on 4D Digital Twins (4DDT), *with CVPR 2026.*
2025 Workshop on Emerging Media Systems, *with SIGCOMM 2025.*
Tutorial on Volumetric Video in the Real World, *with CVPR 2025.*

Student Advising

2026 Xiufeng Huang (Hong Kong Baptist University, NVIDIA Intern)
2025 Yihang Chen (Shanghai Jiao Tong University and Monash University, NVIDIA Intern)
2024 Sharath Girish (University of Maryland, College Park, NVIDIA Intern)
2022-24 Zerui Guo (University of Wisconsin, Madison)
2022-22 Hanchen Li (University of Chicago)
2022 Ali Jahanshahi (UC Riverside, NVIDIA Intern)
Yizhou Chen (University of Wisconsin, Madison)
2017-19 Lucas Cendes (UW CSE)
2017-19 Zachary Calipes (UW EE)
2016 Austin Archiega (UW EE)
Umaymah Khan (UW EE)
2015 Beck Pang (UW EE)
Yufang Sun (UW CSE)

Outreach

2017-19 Chair, *University of Washington CSE Women's Research Day*
2017 Social Media Editor, *ACM SIGARCH*
2015 Co-Chair, *University of Washington CSE Women's Research Day*
2015 Mentor, *ICRA 2015*, Go, Girl Go! Forum
2014 Panelist, *Grace Hopper Celebration of Women in Computing*
2013 Vice President, *Columbia Women in Computer Science*

Talks & Presentations

2022 *Video Scrambling: Fully discarding video contents and generating them on-the-fly*
Workshop on Video Analytics, Stanford University, Invited Talk

2021 *Learning for Better Video Processing Systems*
International Workshop on Performance Analysis of Machine Learning Systems (FastPath), at IEEE ISPASS 2021, Invited Keynote

2021

Hardware-Software Codesign for Visual Computing Systems
UNC Chapel Hill, Invited Talk
Carnegie Mellon University, Invited Talk
Boston College, Invited Talk
San Jose State University, Invited Talk

2020 *A Picture is Worth 1000 Bytes; Everything Else is AI*
Wild and Crazy Ideas (WACI) at ASPLOS, Conference Talk

2019 *Vignette: Perceptual Compression for Video Storage and Processing Systems*
GOMACTech, Conference Talk
ACM Symposium on Cloud Computing, Conference Talk
Asilomar Microcomputer Workshop, Invited Talk
UC Santa Cruz, Invited Talk
UW Photomedia Seminar, Invited Talk

How to throw out 95% of pixels in virtual reality, without anyone noticing!
‼Con West, Conference Talk

2018 *LightDB and Vignette: Database and Storage Systems for Virtual Reality Video*
ASPLUW Retreat, Invited Talk
Google Kirkland, Invited Talk

2017-18 *A Hardware-Friendly Bilateral Solver for Real-Time Virtual Reality Video*
SRC Techcon, Conference Talk
UW Virtual Reality Seminar, Invited Talk
ACM/Eurographics High Performance Graphics, Conference Talk
Oculus Research, Invited Talk

2016-17 *Exploring In-Camera Computation-Communication Tradeoffs*
IEEE International Symposium on Workload Characterization, Conference Talk
UW PLSE Retreat, Invited Talk
UW CSE Industrial Affiliates Annual Meeting, Invited Talk

2014 *CRA-W Panel: Applying to Computer Science Graduate Programs*
Grace Hopper Celebration of Women in Computing

Teaching

2017 Teaching Assistant, *Hardware-Software Interface (CSE 351)*, University of Washington
2016 Teaching Assistant, *Computer Architecture (CSE 548)*, University of Washington
2015 Tutor, *Hardware-Software Interface (CSE 351)*, University of Washington
Tutor, *Programming Languages (CSE 341)*, University of Washington
2014 Instructor, *Emerging Scholars Program (COMS 3998)*, Columbia University
Teaching Assistant, *Embedded Systems Design (COMS 4840)*, Columbia University
2013 Laboratory Assistant, *Introduction to Java*, Columbia Science Honors Program
2012 Teaching Assistant, *Object-Oriented Programming (COMS 1007)*, Columbia University

Previous Employment

2020-2021 **Founder & CEO, Vignette AI**
Founded an AI-based video compression software company.

2020-2021 **CoMotion Postdoctoral Fellow, University of Washington.**
Fellowship to explore the commercialization potential of my research innovations

Research Intern, Facebook Reality Lab.

2018 Design space exploration for deep neural network performance for AR/VR. Manager: Anton Kaplanyan

2017 Explored memory subsystems for AR/VR headsets. Manager: Warren Hunt

ASIC Design Intern, IBM Microelectronics.

2012 Designed and implemented test circuits for fabricated ASIC chips. Manager: Erik Hedberg